Anisotropy of thermal conductivity --- Solution

X23 (2023) — English translation

A1 **Using the least squares method, obtain formulas for finding the coefficients A and B for each temperature value T_0 and for each material. Express your answer in terms of sample average values of products of the form $\Delta T^{k_1} d^{k_2}$, where $k_{1,2}$ are some numbers.** **0.60**

According to the method described in the condition, we compose a system of equations:

$$\begin{cases} \dfrac{\partial S}{\partial A} = 0 \\ \dfrac{\partial S}{\partial B} = 0 \end{cases}, \quad \partial\partial e \quad S = \sum_i \left(\Delta T_{i\,\text{exp}} - \Delta T_{i\,\text{th}}(d_i, A, B)\right)^2 = \sum_i \left(\Delta T_i - Ad_i + BT_0^{1/2} d_i^{3/2}\right)^2$$

Thus, the system will take the form:

$$\begin{cases} \dfrac{\partial}{\partial A} \sum_i \left(\Delta T_i - Ad_i + BT_0^{1/2} d_i^{3/2}\right)^2 = 0 \\ \dfrac{\partial}{\partial B} \sum_i \left(\Delta T_i - Ad_i + BT_0^{1/2} d_i^{3/2}\right)^2 = 0 \end{cases} \Longrightarrow \begin{cases} 0 = \sum_i \left(\Delta T_i d_i - Ad_i^2 + BT_0^{1/2} d_i^{5/2}\right) \\ 0 = \sum_i \left(\Delta T_i d_i^{3/2} - Ad_i^{5/2} + BT_0^{1/2} d_i^3\right) \end{cases}$$

This is a system of linear equations for the coefficients A and B. The coefficients in it are necessary products of the form $\Delta T^{k_1} d^{k_2}$. Using the average values of these products over the experimental sample, this system is written somewhat more elegantly in the form:

$$\begin{cases} 0 = \overline{\Delta T d} - A\overline{d^2} + BT_0^{1/2}\overline{d^{5/2}} \\ 0 = \overline{\Delta T d^{3/2}} - A\overline{d^{5/2}} + BT_0^{1/2}\overline{d^3} \end{cases}$$

Solving this system, we finally obtain:

$$A = \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^{5/2}} - \overline{\Delta T d} \cdot \overline{d^3}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

$$B = \frac{1}{\sqrt{T_0}} \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^2} - \overline{\Delta T d} \cdot \overline{d^{5/2}}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

Answer:

$$A = \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^{5/2}} - \overline{\Delta T d} \cdot \overline{d^3}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

$$B = \frac{1}{\sqrt{T_0}} \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^2} - \overline{\Delta T d} \cdot \overline{d^{5/2}}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

A2 **For molybdenum sulfide (MoS_2), find A and B for each of the temperatures T_0.** **1.20**

$T_0 = 273\ К \overline{\ldots}\ d,\ мкм$ 20 40 60 80 100 120 140 160 180 200 $\Delta T,\ К$ 0.138 0.266 0.394 0.532 0.662 0.777 0.831 0.954 1.137 1.189 0.688 $\Delta Td,\ К\cdot мкм$ 2.76 10.64 23.64 42.56 66.20 93.24 116.34 152.64 204.66 237.80 95.05 $\Delta Td^{3/2},\ К\cdot мкм^{3/2}$ 12.34 6729 18311 380.67 662.00 1021.39 1376.55 1930.76 2745.80 3363.00 1174.29 $d^2,\ мкм^2$ 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2},\ мкм^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 $d^3,\ мкм^3$ 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 323\ К \overline{\ldots}\ d,\ мкм$ 20 40 60 80 100 120 140 160 180 200 $\Delta T,\ К$ 0.137 0.271 0.390 0.496 0.577 0.719 0.788 0.877 1.046 1.098 0.640 $\Delta Td,\ К\cdot мкм$ 2.74 10.84 23.40 39.68 57.70 86.28 110.32 140.32 188.28 219.60 87.92 $\Delta Td^{3/2},\ К\cdot мкм^{3/2}$ 12.25 68.56 181.26 354.91 577.00 945.15 1305.32 1774.92 2526.04 3105.61 1085.10 $d^2,\ мкм^2$ 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2},\ мкм^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 $d^3,\ мкм^3$ 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 373\ К \overline{\ldots}\ d,\ мкм$ 20 40 60 80 100 120 140 160 180 200 $\Delta T,\ К$ 0.139 0.267 0.373 0.483 0.563 0.676 0.760 0.856 0.901 1.023 0.604 $\Delta Td,\ К\cdot мкм$ 2.78 10.68 22.38 38.64 56.30 81.12 106.40 136.96 162.18 204.60 82.20 $\Delta Td^{3/2},\ К\cdot мкм^{3/2}$ 12.43 67.55 173.35 345.61 563.00 888.63 1258.94 1732.42 2175.87 2893.48 1011.13 $d^2,\ мкм^2$ 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2},\ мкм^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 $d^3,\ мкм^3$ 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000

Answer: $T_0,\ К$ 273 323 373 $A,\ \dfrac{К}{мм}$ 7.46 7.18 7.59 $B,\ \dfrac{К^{1/2}}{мм^{3/2}}$ 0.199 0.208 0.298

A3 **Find out whether it is possible to conclude from the results of the experiment that thermal conductivity $\varkappa_\perp$ depends on temperature, underline the correct option on the answer sheet (both sides of the inequality must be explicitly calculated in the solution). Find the arithmetic mean $\overline{A}$ and calculate the thermal conductivity $\varkappa_\perp$ from it.** **0.50**

We substitute the numbers (taking into account that $T_{0\,\max} = 373\ К$, $d_{\max} = 0.2\ мм$):

$$\Delta A < \overline{B} T_{0\,\max}^{1/2} d_{\max}^{1/2} \implies 0.41\ \frac{К}{мм} < 2.03\ \frac{К}{мм}\ -\ \textit{зависит}\ /\ \underline{\textit{не зависит}}.$$

Answer:

$$\overline{A} = 7.41\ \frac{К}{мм}\quad \varkappa_\perp = 53.3\ \frac{мВт}{м\cdot К} \neq \varkappa_\perp(T_0)$$

A4 **Follow steps A2 and A3 for tungsten sulfide (WS_2).** **1.70**

$T_0 = 273\ К \overline{\ldots}\ d,\ мкм$ 20 40 60 80 100 120 140 160 180 200 $\Delta T,\ К$ 0.191 0.381 0.571 0.764 0.946 1.100 1.304 1.443 1.577 1.809 1.009 $\Delta Td,\ К\cdot мкм$ 3.82 15.24 34.26 61.12 94.60 132.00 182.56 230.88 283.86 361.80 140.01 $\Delta Td^{3/2},\ К\cdot мкм^{3/2}$ 17.08 96.39 265.38 546.67 946.00 1445.99 2160.08 2920.43

3808.38 5116.62 1732.30 d^2, $мкм^2$ 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $мкм^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3, $мкм^3$ 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 323\ К \overline{\ldots}$ d, $мкм$ 20 40 60 80 100 120 140 160 180 200 ΔT, $К$ 0.195 0.384 0.566 0.727 0.907 1.083 1.202 1.354 1.518 1.701 0.964 ΔTd, $К \cdot мкм$ 3.9 15.36 33.96 58.16 90.7 129.96 168.28 216.64 273.24 340.2 133.04 $\Delta Td^{3/2}$, $К \cdot мкм^{3/2}$ 17.44 97.15 263.05 520.20 907.00 1423.64 1991.12 2740.30 3665.90 4811.15 1643.70 d^2, $мкм^2$ 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $мкм^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3, $мкм^3$ 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 373\ К \overline{\ldots}$ d, $мкм$ 20 40 60 80 100 120 140 160 180 200 ΔT, $К$ 0.201 0.367 0.549 0.733 0.877 1.037 1.172 1.330 1.439 1.629 0.933 ΔTd, $К \cdot мкм$ 4.02 14.68 32.94 58.64 87.70 124.44 164.08 212.80 259.02 325.80 128.41 $\Delta Td^{3/2}$, $К \cdot мкм^{3/2}$ 17.98 92.84 255.15 524.49 877.00 1363.17 1941.42 2691.73 3475.12 4607.51 1584.64 d^2, $мкм^2$ 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $мкм^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3, $мкм^3$ 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 Substitute the numbers (taking into account that $T_{0\,\max} = 373\ К$, $d_{\max} = 0.2\ мм$):

$$\Delta A < \overline{B} T_{0\,\max}^{1/2} d_{\max}^{1/2} \implies 0.23\ \frac{К}{мм} < 2.17\ \frac{К}{мм} \quad - \ \textit{зависит} \ / \ \underline{\textit{не зависит}}$$

We substitute the numbers (taking into account that $T_{0\,\max} = 373\ К$, $d_{\max} = 0.2\ мм$):

$$\Delta A < \overline{B} T_{0\,\max}^{1/2} d_{\max}^{1/2} \implies 0.23\ \frac{К}{мм} < 2.17\ \frac{К}{мм} \quad - \ \textit{зависит} \ / \ \underline{\textit{не зависит}}$$

$$\overline{A} = 10.47\ \frac{К}{мм} \varkappa_\perp = 37.7\ \frac{мВт}{м \cdot К} \neq \varkappa_\perp(T_0)$$

Answer: T_0, $К$ 273 323 373 A, $\frac{К}{мм}$ 10.37 10.44 10.60 B, $\frac{К^{1/2}}{мм^{3/2}}$ 0.197 0.256 0.300

$$\overline{A} = 10.47\ \frac{К}{мм} \varkappa_\perp = 37.7\ \frac{мВт}{м \cdot К} \neq \varkappa_\perp(T_0)$$

B1 **Using the least squares method, obtain a formula for finding the coefficient A for each temperature value T_0 and for each material. Express your answer in terms of sample average values of products of the form $\Delta T^{k_1} d^{k_2}$, where $k_{1,2}$ are some numbers.** **0.60**

According to the method described in the condition, we compose a system of equations:

$$\begin{cases} \dfrac{\partial S}{\partial A} = 0 \\ \dfrac{\partial S}{\partial B} = 0 \end{cases}, \quad где \quad S = \sum_i \left(\Delta T_{i\,\exp} - \Delta T_{i\,\text{th}}(d_i, A, B)\right)^2 = \sum_i \left(\Delta T_i - \frac{A}{d_i} + \frac{B}{d_i^2}\right)^2$$

Thus, the system will take the form:

$$\begin{cases} \dfrac{\partial}{\partial A}\sum_i \left(\Delta T_i - \dfrac{A}{d_i} + \dfrac{B}{d_i^2}\right)^2 = 0 \\ \dfrac{\partial}{\partial B}\sum_i \left(\Delta T_i - \dfrac{A}{d_i} + \dfrac{B}{d_i^2}\right)^2 = 0 \end{cases} \Longrightarrow \begin{cases} 0 = \sum_i \left(\Delta T_i d_i^{-1} - A d_i^{-2} + B d_i^{-3}\right) \\ 0 = \sum_i \left(\Delta T_i d_i^{-2} - A d_i^{-3} + B d_i^{-4}\right) \end{cases}$$

This is a system of linear equations for the coefficients A and B. The coefficients in it are necessary products of the form $\Delta T^{k_1} d^{k_2}$. Using the average values of these products over the experimental sample, this system is written somewhat more elegantly in the form:

$$\begin{cases} 0 = \overline{\Delta T d^{-1}} - A\overline{d^{-2}} + B\overline{d^{-3}} \\ 0 = \overline{\Delta T d^{-2}} - A\overline{d^{-3}} + B\overline{d^{-4}} \end{cases}$$

Solving this system for A, we finally obtain:

$$A = \frac{\overline{\Delta T d^{-2}} \cdot \overline{d^{-3}} - \overline{\Delta T d^{-1}} \cdot \overline{d^{-4}}}{\overline{d^{-3}}^2 - \overline{d^{-2}} \cdot \overline{d^{-4}}}.$$

Answer:

$$A = \frac{\overline{\Delta T d^{-2}} \cdot \overline{d^{-3}} - \overline{\Delta T d^{-1}} \cdot \overline{d^{-4}}}{\overline{d^{-3}}^2 - \overline{d^{-2}} \cdot \overline{d^{-4}}}$$

B2 **For molybdenum sulfide (MoS_2), find A for each of the temperatures T_0. From here, for each T_0, find the values of $\varkappa_\parallel$. Find the thermal conductivity anisotropy coefficient $\rho = \dfrac{\varkappa_\parallel}{\varkappa_\perp}$ at temperature $T_0 = 273\ K$.** **1.00**

$T_0 = 273\ K\ \overline{\dots}\ d,\ мкм$ 120 140 160 180 200 $\Delta T,\ K$ 1.372 1.157 1.055 0.948 0.846 $\Delta T d^{-1},\ 10^{-3}\ K \cdot мкм^{-1}$ 11.43 8.26 6.59 5.27 4.23 7.16 $\Delta T d^{-2},\ 10^{-6}\ K \cdot мкм^{-2}$ 95.28 59.03 41.21 29.26 21.15 49.19 $d^{-2},\ 10^{-6}\ мкм^{-2}$ 69.44 51.02 39.06 30.86 25.00 43.08 $d^{-3},\ 10^{-9}\ мкм^{-3}$ 578.70 364.43 244.14 171.47 125.00 296.75 $d^{-4},\ 10^{-9}\ мкм^{-4}$ 4.823 2.603 1.526 0.953 0.625 2.106
$T_0 = 323\ K\ \overline{\dots}\ d,\ мкм$ 120 140 160 180 200 $\Delta T,\ K$ 1.407 1.247 1.144 1.019 0.948 $\Delta T d^{-1},\ 10^{-3}\ K \cdot мкм^{-1}$ 11.73 8.91 7.15 5.66 4.74 7.64 $\Delta T d^{-2},\ 10^{-6}\ K \cdot мкм^{-2}$ 97.71 63.62 44.69 31.45 23.70 52.23 $d^{-2},\ 10^{-6}\ мкм^{-2}$ 69.44 51.02 39.06 30.86 25.00 43.08 $d^{-3},\ 10^{-9}\ мкм^{-3}$ 578.70 364.43 244.14 171.47 125.00 296.75 $d^{-4},\ 10^{-9}\ мкм^{-4}$ 4.823 2.603 1.526 0.953 0.625 2.106
$T_0 = 373\ K\ \overline{\dots}\ d,\ мкм$ 120 140 160 180 200 $\Delta T,\ K$ 1.455 1.356 1.194 1.107 1.024 $\Delta T d^{-1},\ 10^{-3}\ K \cdot мкм^{-1}$ 12.13 9.69 7.46 6.15 5.12 8.11 $\Delta T d^{-2},\ 10^{-6}\ K \cdot мкм^{-2}$ 101.04 69.18 46.64 34.17 25.60 55.33 $d^{-2},\ 10^{-6}\ мкм^{-2}$ 69.44 51.02 39.06 30.86 25.00 43.08 $d^{-3},\ 10^{-9}\ мкм^{-3}$ 578.70 364.43 244.14 171.47 125.00 296.75 $d^{-4},\ 10^{-9}\ мкм^{-4}$ 4.823 2.603 1.526 0.953 0.625 2.106

$$\rho = 9.85 \cdot 10^2$$

Answer: $T_0,\ K$ 273 323 373 $A,\ мкм \cdot K$ 179.6 218.8 247.5 $\varkappa_\parallel,\ \dfrac{Вт}{м \cdot K}$ 52.5 43.1 38.1

$$\rho = 9.85 \cdot 10^2$$

B3 **Find n.** **0.40**

You can calculate coefficients $\varkappa_{\|,0}$ and n using least squares for a modeling function of the form:

$$\varkappa_{\|} = \frac{\varkappa_{\|,0}}{T_0^n}$$

using a standard calculator.

Answer:

$$n = 1.03$$

B4 **Follow steps B2 and B3 for tungsten sulfide $(\mathrm{WS_2})$.** **1.40**

$T_0 = 273\ K \overline{\ldots}\ d,\ мкм$ 120 140 160 180 200 160 $\Delta T,\ K$ 1.596 1.417 1.284 1.163 1.084 1.309 $\Delta T d^{-1},\ 10^{-3}\ K \cdot мкм^{-1}$ 13.30 10.12 8.03 6.46 5.42 8.67 $\Delta T d^{-2},\ 10^{-6}\ K \cdot мкм^{-2}$ 110.83 72.30 50.16 35.90 27.10 59.26 $d^{-2},\ 10^{-6}\ мкм^{-2}$ 69.44 51.02 39.06 30.86 25.00 43.08 $d^{-3},\ 10^{-9}\ мкм^{-3}$ 578.70 364.43 244.14 171.47 125.00 296.75 $d^{-4},\ 10^{-9}\ мкм^{-4}$ 4.823 2.603 1.526 0.953 0.625 2.106
$T_0 = 323\ K \overline{\ldots}\ d,\ мкм$ 120 140 160 180 200 160 $\Delta T,\ K$ 1.678 1.482 1.370 1.255 1.162 1.389 $\Delta T d^{-1},\ 10^{-3}\ K \cdot мкм^{-1}$ 13.98 10.59 8.56 6.97 5.81 9.18 $\Delta T d^{-2},\ 10^{-6}\ K \cdot мкм^{-2}$ 116.53 75.61 53.52 38.73 29.05 62.69 $d^{-2},\ 10^{-6}\ мкм^{-2}$ 69.44 51.02 39.06 30.86 25.00 43.08 $d^{-3},\ 10^{-9}\ мкм^{-3}$ 578.70 364.43 244.14 171.47 125.00 296.75 $d^{-4},\ 10^{-9}\ мкм^{-4}$ 4.823 2.603 1.526 0.953 0.625 2.106
$T_0 = 373\ K \overline{\ldots}\ d,\ мкм$ 120 140 160 180 200 160 $\Delta T,\ K$ 1.763 1.607 1.509 1.355 1.307 1.508 $\Delta T d^{-1},\ 10^{-3}\ K \cdot мкм^{-1}$ 14.69 11.48 9.43 7.53 6.54 9.93 $\Delta T d^{-2},\ 10^{-6}\ K \cdot мкм^{-2}$ 122.43 81.99 58.95 41.82 32.68 67.57 $d^{-2},\ 10^{-6}\ мкм^{-2}$ 69.44 51.02 39.06 30.86 25.00 43.08 $d^{-3},\ 10^{-9}\ мкм^{-3}$ 578.70 364.43 244.14 171.47 125.00 296.75 $d^{-4},\ 10^{-9}\ мкм^{-4}$ 4.823 2.603 1.526 0.953 0.625 2.106 You can calculate coefficients $\varkappa_{\|,0}$ and n using least squares for a modeling function of the form:

$$\varkappa_{\|} = \frac{\varkappa_{\|,0}}{T_0^n}$$

using a standard calculator.

You can calculate coefficients $\varkappa_{\|,0}$ and n using least squares for a modeling function of the form:

$$\varkappa_{\|} = \frac{\varkappa_{\|,0}}{T_0^n}$$

using a standard calculator.

$$\rho = 1.000 \cdot 10^3 n = 1.02$$

Answer: $T_0,\ K$ 273 323 373 $A,\ мкм \cdot K$ 250.0 276.7 325.7 $\varkappa_{\|},\ \dfrac{Вт}{м \cdot K}$ 37.7 34.1 29.0

$$\rho = 1.000 \cdot 10^3 n = 1.02$$

B5 **Using the least squares method, obtain a formula for finding the coefficient $\Delta T_{\max}$ for each temperature value T_0 and for each material. Express your answer in terms of sample average values of products of the form $\Delta T^{k_1} d^{k_2}$, where $k_{1,2}$ are some numbers.** **0.60**

According to the method described in the condition, we compose a system of equations:

$$\begin{cases} \dfrac{\partial S}{\partial(\Delta T_{\max})} = 0 \\ \dfrac{\partial S}{\partial C} = 0 \end{cases}, \quad где \quad S = \sum_i \left(\Delta T_{i\,\exp} - \Delta T_{i\,\mathrm{th}}(d_i, A, B)\right)^2 = \sum_i \left(\Delta T_i - \Delta T_{\max} + C d_i\right)^2$$

Thus, the system will take the form:

$$\begin{cases} \dfrac{\partial}{\partial(\Delta T_{\max})} \sum_i \left(\Delta T_i - \Delta T_{\max} + C d_i\right)^2 = 0 \\ \dfrac{\partial}{\partial C} \sum_i \left(\Delta T_i - \Delta T_{\max} + C d_i\right)^2 = 0 \end{cases} \implies \begin{cases} 0 = \sum_i \left(\Delta T_i - \Delta T_{\max} + C d_i\right) \\ 0 = \sum_i \left(\Delta T_i d_i - \Delta T_{\max} d_i + C d_i^2\right) \end{cases}$$

This is a system of linear equations for the coefficients A and B. The coefficients in it are necessary products of the form $\Delta T^{k_1} d^{k_2}$. Using the average values of these products over the experimental sample, this system is written somewhat more elegantly in the form:

$$\begin{cases} 0 = \overline{\Delta T} - \Delta T_{\max} + C\bar{d} \\ 0 = \overline{\Delta T d} - \Delta T_{\max}\bar{d} + C\overline{d^2} \end{cases}$$

Solving this system for $\Delta T_{\max}$, we finally obtain:

$$\Delta T_{\max} = \frac{\overline{\Delta T} \cdot \overline{d^2} - \bar{d} \cdot \overline{\Delta T d}}{\overline{d^2} - \bar{d}^2}.$$

Answer:

$$\Delta T_{\max} = \frac{\overline{\Delta T} \cdot \overline{d^2} - \bar{d} \cdot \overline{\Delta T d}}{\overline{d^2} - \bar{d}^2}$$

B6 **For each of the materials, find $\Delta T_{\max}$ for each of the temperatures T_0.** **1.20**

You can, of course, use the formula from the previous paragraph, but since the modeling function is simply a linear relationship, the value $\Delta T_{\max}$ can be calculated using least squares and using a standard calculator.

Answer: $T_0,\ K$ 273 323 373 $\overset{MoS_2:}{\Delta T_{\max}},\ K$ 2.841 2.451 2.159 $\overset{WS_2:}{\Delta T_{\max}},\ K$ 3.155 2.679 2.293

B7 **For each of the materials, find m. What makes the main contribution to heat transfer between the sample and the environment – thermal conductivity or radiation? Underline the correct option on your answer sheet.** **0.80**

You can calculate coefficients $\Delta T_{\max,0}$ and m using least squares for a modeling function of the form:

$$\Delta T_{\max} = \frac{\Delta T_{\max,0}}{T_0^m}$$

using a standard calculator. As a result, we obtain: for MoS_2, $m = 0.88$ – the main contribution is made by *теплопроводность* / radiation; for WS_2, $m = 0.83$ – the main contribution is made by *теплопроводность* / radiation.

You can calculate coefficients $\Delta T_{\max,0}$ and m using least squares for a modeling function of the form:

$$\Delta T_{\max} = \frac{\Delta T_{\max,0}}{T_0^m}$$

using a standard calculator. As a result we get:

The main contribution to heat transfer between the sample and the environment is thermal conductivity.

Answer: Material MoS_2 WS_2 m 0.88 0.83 Thermal conductivity makes the main contribution to the heat transfer between the sample and the environment.